Practical - 2

Practical Title: Semantic HTML5 Pages with Accessibility-Ready Structure

Problem Definition: Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure.

A. Theory

1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

   Answer: Yes, Here it's a short descriptive part on it

- **<header>**: Contains introductory content, logos, navigation, or search bars for a page or section.
- **<nav>**: Wraps the primary navigation links of the website.
- **<main>**: Holds the unique, dominant content of the page. Only use one per page.
- **<section>**: Groups related content together under a single theme, usually requiring a heading (<h2>-<h6>).
- **<article>**: Encloses self-contained content that makes sense on its own (e.g., blog posts, product cards).
- **<aside>**: Holds indirectly related content, like sidebars, callout boxes, or advertising.
- **<footer>**: Contains closing information, such as copyright data, contact info, or secondary links.

https://dev.to/sharique_siddiqui_8242dad/semantic-html-header-footer-nav-section-article-aside-main-2480

2. Are form labels, headings, image alt text, and navigation links accessible?

Answer:

Yes, form labels, headings, image alt text, and navigation links are highly accessible when built using semantic HTML. They serve as the core foundational pillars of web accessibility, allowing users with visual, cognitive, or physical impairments to successfully navigate digital spaces using assistive technologies like screen readers.

3. Is the page structure consistent across all pages?

Answer:

Yes, my page structure is completely consistent across all pages**.** Every page uses the exact same header, main content layout, navigation structure, and footer to ensure a predictable and fully accessible user experience .

B. Result **(Keep only one option which is suitable to practical)**

The required StudentHub pages were successfully created using semantic HTML5 elements and an accessibility-ready structure. The pages provide a consistent navigation and content structure across the portal.

C. Evidence of Work (Screenshots)

D. GitHub Repository Link:

https://github.com/Dhanraj2408/StudentHub

E. Challenges Faced:

During the implementation of the StudentHub pages, the following challenges were faced:

- Understanding the correct use of semantic HTML5 tags instead of using elements for everything.
- Maintaining a consistent structure and navigation across multiple pages.
- Creating proper form labels and accessible input fields.
- Maintaining a logical heading hierarchy on different pages.

- Adding meaningful alternative text to images.
- Managing common navigation and footer elements across all pages.
- Ensuring that the HTML structure remains clean and easy to maintain.

These challenges were solved by referring to HTML5 semantic elements, following accessibility practices, and maintaining a common page structure across the StudentHub portal.

F. Learning Outcomes:

After completing this practical, I learned:

- The purpose and proper usage of semantic HTML5 elements.
- How to structure webpages using <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>.
- How to create accessibility-friendly forms using labels and appropriate input elements.
- The importance of a logical heading hierarchy.
- How to use descriptive alt text for images.
- How to create consistent navigation across multiple webpages.
- How semantic HTML improves website structure, readability, and accessibility.
- How to organize multiple static HTML pages for a larger web project.

Faculty Remarks: